

CS329Z: Compound AI Systems and Agents

Stanford Frontier AI Learning System

Baseline: October 6, 2026. Print companion to the HTML course site:
course map, prerequisites, notation and shapes, glossary,
cheatsheet, crash course.

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Course map

Baseline 2026-10-06. Build 2026-10-07. Units U01-U04 built by the first builder. Units U05-U08 plus consolidation built by the second builder (2026-10-07). S05-S20 remain PLANNED / SOURCE ATTRIBUTION PENDING.

Official schedule (verified 2026-10-07 from <https://cs329z.stanford.edu/>)

Classes meet Mondays and Wednesdays, 1:30-2:50 p.m. Pacific, Skilling Auditorium.

Session	Date (2026)	Title	Unit map	Claim class
S01	Wed 23 Sep	Foundations session: what are agentic systems, the spectrum from monolithic models to compound AI systems to agents, the three engineering challenges (decomposition, data, evaluation)	U01-C01, U01-C02, U01-C03	Released, schedule inspected in-
S02	Mon 28 Sep	LLMs for Builders: decoding strategies, transformer attention (including linear and hybrid), context length, pretraining-to-post-training pipeline, inference and test-time scaling, structured I/O and constrained decoding, context engineering	U01-C04 through U01-C10	Released, schedule inspected in-
S03	Wed 30 Sep	Building Blocks RAG: grounding and hallucination, embeddings and vector stores, chunking strategies, hybrid search, cross-encoders and late interaction (ColBERT)	U02 (all)	Released, schedule inspected in-
S04	Mon 5 Oct	Tool Use and Function Calling: the REPL, function-calling APIs, MCP, designing good tools, code-execution sandboxes, error handling and retries	U03 (all)	Released, schedule inspected in-

Session	Date (2026)	Title	Unit map	Claim class
S05	Wed 7 Oct	Frameworks and Orchestration: DSPy (signatures, modules, optimizers), LangChain/LangGraph, LlamaIndex. Abstraction levels	U04-C01, U04-C02	Planned
S06	Mon 12 Oct	Agent Design Patterns and Scaffolds: workflows-vs-agents taxonomy, five composable workflow patterns, agent patterns (ReAct, plan-and-execute, reflection)	U04-C03, U04-C04, U04-C05, U04-C11, U04-C12	Planned
S07	Wed 14 Oct	Agent Memory Architectures: short vs long-term memory, memory as tool-based actions, file system as externalized memory, structured memory paradigms, cross-agent memory	U04-C06, U04-C07, U04-C09	Planned
S08	Mon 19 Oct	Multi-Agent Systems: single vs multi-agent architectures, orchestration, handoffs and state transfer, delegation, coordination, error propagation	U04-C08, U04-C09, U04-C10, U04-C11	Planned

Session	Date (2026)	Title	Unit map	Claim class
S09	Wed 21 Oct	Optimization: prompts to fine-tuning, GEPA/MIPROv2/OPRO/TextGrad, LoRA/QLoRA, distillation, RLHF/DPO	U05	Planned
S10	Mon 26 Oct	Guest lecture (TBA)	Unmapped	Planned
S11	Wed 28 Oct	Data for Agentic Systems: traces, demonstrations, feedback, data fly-wheels, synthetic data	U05	Planned
S12	Mon 2 Nov	Data Selection and Quality: informative data, filtering, tiny benchmarks, annotation	U05	Planned

Session	Date (2026)	Title	Unit map	Claim class
S13	Wed 4 Nov	Evaluation Fundamentals and Benchmark Design: the 4-tuple (request, environment, stopping criteria, scorer), benchmark properties, scaffolding	U06	Planned
S14	Mon 9 Nov	LLM-as-Judge and Evaluation Infrastructure: grader types, judge prompts, biases, point vs pairwise, pass@k vs pass^k, test-rig design	U06	Planned
S15	Wed 11 Nov	Agent Safety and Guardrails: privacy, prompt injection, red-teaming, sandboxing, permissions, approvals	U07	Planned

Session	Date (2026)	Title	Unit map	Claim class
S16	Mon 16 Nov	Guest lecture (TBA)	Unmapped	Planned
S17	Wed 18 Nov	Coding Agents and Proactive Agents: coding agent architectures, SWE- bench, scaffolds	U07	Planned
S18	Mon 23 Nov	No class (Thanksgiving recess)	-	-

Session	Date (2026)	Title	Unit map	Claim class
S19	Mon 30 Nov	Proactive Agents: user models, next action prediction, mixed initiative, privacy and trust	U07	Planned
S20	Wed 2 Dec	Frontiers and Open Problems: multimodal, web/computer use, science agents, long-running architectures, observability, reliability, scalability, interpretability	U08	Planned
Finals	Dec 7- 11	Final project demo day	U08	Planned

Concept-to-session provenance in this build

Note: the official S01 title carries one extra word after "Foundations and". That word is omitted from this build per ASD-STE100.

- U01: C01-C03 map to S01 titles, C04-C10 map to S02 titles. Source-supported at schedule-title level. C11 (simple baseline) and C12 (architecture choice) are requested extensions taught as independent theory, labeled PLANNED / SOURCE ATTRIBUTION PENDING.
- U02: C01, C02, C03, C04, C05, C07, C08 map to S03 titles. Source-supported at schedule-title level. C06 (metadata/permissions), C09 (rerank), C10 (citations), C11 (insufficient evidence), C12 (retrieval diagnosis) are requested extensions taught as independent theory, labeled pending.
- U03: C01, C02, C04, C06, C08, C12 map to S04 titles. Source-supported at schedule-title level. C03 (tool/result loop), C05 (authentication/authorization), C07 (timeouts), C09 (idempotency), C10 (approval), C11 (output contracts) are requested extensions taught as independent theory, labeled pending.
- U04: S05-S08 are planned (S05 not yet held at build time). All twelve concepts taught as independent theory, labeled PLANNED / SOURCE ATTRIBUTION PENDING. U04-C01 is additionally grounded in the DSPy paper abstract (SRC-06), U04-C03, U04-C04 in the Anthropic article (SRC-04), U03-C04 in the MCP spec (SRC-05).

Assessment on the official page (learning objectives, not reproduced work)

Two fully applied homework assignments: HW1 released Oct 5 (GitHub), due Oct 30, HW2 released Oct 26, due Nov 20. Quarter-long project: proposal Oct 9, midpoint demo video Nov 4, midway report Nov 6, paper video Nov 13, peer reviews Nov 30, final report and demo Dec 7-11. This course builds original equivalent practice only. It never solves or reproduces assessed work.

Prerequisites

Shared bridges P01-P24 live at [workspace/stanford-frontier-ai/v2-pack/shared/prerequisites/](#). Link, do not rebuild. Each unit below names its parent-level prerequisites from the inventory and the local remediation this course adds. Every lesson also opens with a local self-contained remediation section.

Prerequisite graph for U01-U08

Unit	Inventory prerequisites	Bridge content	Shared bridge files
U01	P10, P14, P20	ML foundations and evaluation. Transformer mechanics. Tools, APIs, and agent state	p10_ml_foundations.md, p14_transformer.md, p20_tools.md
U02	P19, P21	Retrieval and information access. Security, privacy, and safety	p19_retrieval.md, p21_security.md
U03	P16, P20, P21	Distributed systems and networking. Tools, APIs, and agent state. Security, privacy, and safety	p16_distributed.md, p20_tools.md, p21_security.md
U04	P16, P20	Distributed systems and networking. Tools, APIs, and agent state	p16_distributed.md, p20_tools.md

Unit	Inventory prerequi- sites	Bridge content	Shared bridge files
U05	P09, P14, P17, P22	Optimization. Transformer mechanics. Reinforcement learning. Experiments	p09_optimization.md, p14_ transformer.md, p17_rl.md, p22_experiments.md
U06	P07, P10, P22	Statistical estimation. ML foundations and evaluation. Experiments	p07_estimation.md, p10_ml_ foundations.md, p22_experiments.md
U07	P20, P21, P24	Tools, APIs, and agent state. Se- curity, privacy, and safety. Pro- duction ML (P24 is a composite of P10/P16/P21/P22/P23, no dedicated file)	p20_tools.md, p21_security.md
U08	P16, P22, P24	Distributed systems and networking. Experiments. Production ML and eco- nomics (P24 is a composite, no dedi- cated file)	p16_distributed.md, p22_experiments. md, p23_economics.md

Deep chain used inside lessons: P01 numeracy, P02 Python, P03 vectors, P06 probability, P07 estimation, P08 information, P10 ML foundations, P13 language, P14 transformer, P15 hardware, P16 distributed, P20 tools, P21 security, P22 experiments.

Local remediation per unit

- U01: Bernoulli and softmax refresher (logits, temperature, normalization). Attention cost counting ($O(n^2 d)$ with n context length, d model width). Train versus inference. Likelihood as a training objective. Taught inside `lessons/u01_lesson.md`.
- U02: cosine/dot/L2 refresher with a 2D toy, BM25 intuition (term frequency saturation, inverse document frequency). Precision and recall for retrieval. Ranking as sorting by score. Taught inside `lessons/u02_lesson.md`.
- U03: JSON and HTTP refresher. State machines (states, transitions, halting). Exponential backoff arithmetic. Authentication versus authorization split. Taught inside `lessons/u03_lesson.md`.
- U04: graphs and DAGs refresher.
- U05: averages as expectations, parameter counting, matrix rank, preference pairs, overfitting. Taught inside `lessons/u05_lesson.md`.

- U06: means, standard errors, correlations, win rates, nondeterminism. Taught inside lessons/u06_lesson.md.
- U07: trust boundaries, instruction hierarchy, capabilities, consent. Taught inside lessons/u07_lesson.md.
- U08: action spaces, spans, percentiles, ownership. Taught inside lessons/u08_lesson.md. The LLM call as a pure function of (prompt, parameters). Summary as lossy compression. Fixed-point iteration intuition for loops. Taught inside lessons/u04_lesson.md.

Diagnostic (closed book, 20 minutes)

Answer in diagnostics/diagnostic_keys.md, kept separate.

D1. Write the softmax of logits [1.0, 2.0, 3.0] at temperature 1. Give each probability to two decimals. D2. A transformer reads 4096 tokens with model width 1024. Estimate the attention FLOPs up to a constant factor. D3. Two unit vectors have dot product 0.8. What is the angle between them in degrees, roughly? D4. Define precision and recall for a retrieval system in one sentence each. D5. Name the three MCP participant roles and the two transport mechanisms. D6. A tool call fails with probability 0.2 per try, independently. After 3 tries with no backoff, what is the probability all three fail? D7. Sketch the ReAct loop as three boxes with arrows. Label the arrows. D8. What breaks first when an agent loop has no stopping rule? Name one concrete failure.

How to use the diagnostic

Score each item 0/1. A score below 5 means: read the local remediation of the failed unit first, then the shared bridge, then the lesson. A score of 5 or above means: start the lessons and use remediation on demand. Do not infer mastery from reading. The diagnostic measures entry state only.

Notation and shapes

One meaning per symbol across U01-U04. Shapes use (rows, cols) or named axes. Scalars are lowercase, vectors bold lowercase, matrices bold uppercase.

General

Symbol	Meaning	Units/shape
x	input text or token sequence	sequence of token ids
y	model output text or token sequence	sequence of token ids
z	logits over the vocabulary	R^V
V	vocabulary size	count, e.g. 32000

Symbol	Meaning	Units/shape
T	temperature	positive scalar, dimensionless
p _i	probability of token i	dimensionless, sums to 1
n	context length in tokens	count
d	model width (embedding dimension)	count

Symbol	Meaning	Units/shape
L	number of transformer layers	count
h	number of attention heads	count
theta	model parameters	parameter vector
D	dataset or document collection	set

Symbol	Meaning	Units/shape
q	query (text or vector)	text. Or $\mathbb{R}^d$ as vector
d_i	document i (U02 uses d for document, not width. Lesson dis-ambiguates)	text. Or $\mathbb{R}^d$ as vector
s(q, d)	retrieval score of document d for query q	scalar

Attention (U01)

Q, K, V: query, key, value matrices, each (n, d_k) per head with $d_k = d / h$. $\text{Attention}(Q, K, V) = \text{softmax}(Q K^T / \sqrt{d_k}) V$, output (n, d_k) per head. KV cache: stored K and V for past tokens, shape $(L, 2, n, d)$ across the model.

Retrieval (U02)

$e(x)$: embedding of text x, $\mathbb{R}^d$. $\cos(a, b) = (a \cdot b) / (||a|| ||b||)$, dimensionless in $[-1, 1]$. Top-k: the k documents with the largest $s(q, d)$. BM25(q, d): scalar lexical score from term frequency and inverse document frequency.

Tools and protocols (U03)

f: a tool, a named function with a schema. args: tool arguments, a JSON object. obs: tool result (observation), JSON or text. s_t: agent state at step t. a_t: action at step t (tool call or final answer). pi: policy mapping state to action distribution. tau: timeout in seconds. R: retry count (integer).

Frameworks and memory (U04)

S: DSPy signature, an input-output contract. M: DSPy module, a parameterized program step. Phi: compiled program after optimization. G: workflow graph (nodes = calls, edges = data flow). H_t: conversation history up to step t. mem_S: short-term memory (working context). Mem_L: long-term memory (persistent store). B: step or token budget, integer.

Probability and evaluation

$P(A)$: probability of event A . $E[X]$: expectation. $\text{Var}(X)$: variance. $\text{pass}@k$: probability that at least one of k samples passes, given per-sample pass rate p : $1 - (1 - p)^k$ under independence.

Glossary

One meaning per term. First use in lessons repeats the short definition.

- **Acceptance gate:** a bar with a measurement that a system must clear before the next rollout step.
- **Agent:** a system in which an LLM dynamically directs its own process and tool use, maintaining control over how it completes a task (Anthropic 2024 usage).
- **Agentic system:** the umbrella term covering workflows and agents.
- **Attack success rate (ASR):** fraction of red-team attacks that achieve the attacker's goal.
- **Augmented LLM:** an LLM with retrieval, tools, and memory attached (SRC-04).
- **Baseline:** the simplest system worth comparing against, usually a single LLM call or a from-scratch pipeline with no tuning.
- **Bradley-Terry:** a model turning pairwise win counts into one strength number per system.
- **Calibration gap:** mean validator score minus mean human label on a calibration set.
- **Capability:** a named permission (read, write, send) enforced at the action layer.
- **Checkpoint:** saved run state that a long run resumes from after a crash.
- **Chunk:** a contiguous slice of a document fed to an embedder or to the model.
- **ColBERT:** late-interaction retrieval. Token-level embeddings compared with a max-sim operator at query time.
- **Compound AI system:** a system of multiple interacting components (LLMs, retrievers, tools, optimizers) built to serve people (course framing, S01).
- **Consent gradient:** consent levels (implicit, explicit-once, explicit-each-time) matched to action stakes.
- **Constrained decoding:** sampling restricted to token sequences that satisfy a grammar or schema.
- **Context engineering:** choosing what goes into the context window (retrieved text, history, instructions) and in what form.
- **Cross-encoder:** a model that reads query and document together and outputs one relevance score.

- Data flywheel: the deploy-collect-annotate-train-redeploy loop that compounds deployment data.
- Decomposition: splitting a task into subtasks with defined inputs, outputs, and checks.
- Distillation: training a small student on a large teacher's labels (hard or soft targets).
- DPO: direct preference optimization. Trains the policy on (x, y_w, y_l) pairs with $-\log \sigma(\beta m)$, no RL stage.
- DSPy: a programming model for LM pipelines with declarative signatures, parameterized modules, and a compiler (optimizer) that maximizes a metric.
- Embedding: a vector in $\mathbb{R}^d$ that represents text for similarity search.
- Eval tuple: the four parts that make a benchmark reproducible (request, environment, stopping, scorer).
- Evaluator-optimizer: a workflow in which one LLM call generates and another critiques in a loop.
- Grounding: tying a generated claim to retrieved evidence. Not the same as truth.
- Hallucination: a fluent, confident statement not supported by evidence or fact.
- Holdout: an eval set kept from training. The sealed tier opens once.
- Human alignment: the correlation between a benchmark metric and human-judged quality.
- Hybrid search: a retrieval score that mixes lexical (BM25) and dense (embedding) signals.
- Idempotency key: a token deduping retried requests so repeats have one effect.
- Idempotency: the property that repeating a request has the same effect as doing it once.
- Injection (prompt): an instruction smuggled in user text (direct) or tool/file/web content (indirect).
- Instruction hierarchy: system outranks developer outranks user outranks tool output.
- Late interaction: scoring that keeps per-token vectors and combines them at query time instead of compressing to one vector.
- Least privilege: granting the agent the smallest capability set that completes the task.
- LoRA: low-rank adaptation. Trains BA (rank r) while the base matrix stays frozen. QLoRA adds 4-bit base storage.

- MCP: Model Context Protocol. Host/client/server roles, JSON-RPC 2.0 data layer, stdio and Streamable HTTP transports.
- Milestone: a verifiable subgoal in a long-horizon plan.
- Mixed initiative: control passing between agent and user by confidence and stakes.
- Modality ablation: toggling input modalities to price what each buys.
- Orchestrator-workers: a workflow in which a central LLM splits the task, delegates to workers, and synthesizes results.
- p99: the cost or latency that only 1 percent of tasks exceed.
- Parallelization: a workflow that runs subtasks or repeated attempts at once, then aggregates.
- Pass@k: probability that at least one of k independent samples passes.
- Pass^k: probability that all k samples pass (reliability).
- Permission boundary: the enforced list of what the agent may do.
- Prompt chaining: a workflow that decomposes a task into a fixed sequence of LLM calls.
- QLoRA: LoRA with the frozen base in 4-bit, fitting large models on small GPUs.
- ReAct: an agent pattern that interleaves reasoning traces with actions and observations.
- Recall: fraction of relevant items retrieved. Precision: fraction of retrieved items relevant.
- Recovery ladder: retry, rollback, escalate, compensate, cheapest first.
- Red-team: adversarial testing that invents attacks and measures the attack success rate.
- REPL: read-eval-print loop. An interactive execution environment, here the model for code-execution tools.
- Reranker: a second-stage scorer applied to a small candidate set from first-stage retrieval.
- RLHF: reinforcement learning from human feedback. Reward model plus policy optimization. DPO replaces both stages.
- Routing: a workflow that classifies an input and sends it to a specialized downstream path.
- Runbook: the written incident response, rehearsed by drill.
- Sandbox: an isolated execution environment with limited permissions and no unmediated side effects.

- Scaffold: the code around the model (loops, prompts, tools, checks) that turns calls into a system.
- Scaffolding fidelity: how closely benchmark tools match production tools.
- Span: one logged unit of work with tokens and cost. A trace is a tree of spans.
- Stopping rule: a condition that ends an agent loop (answer found, budget spent, approval denied).
- Structured I/O: model input and output constrained to a machine-readable schema such as JSON.
- Synthetic data: model-generated training examples filtered by a verifier.
- Temperature: the scalar T that divides logits before softmax, $T \rightarrow 0$ approaches greedy, large T flattens the distribution.
- Test-time scaling: spending more compute at inference (more samples, longer chains, more search) to raise quality.
- Tool: a named function with a declared schema that the model can call. Execution happens outside the model.
- Trace (agent): the thought/action/observation sequence. Trace (production): the span tree of a run.
- User model: the agent's stored picture of the user's goals, preferences, and constraints.
- Validator: a scoring function an optimizer trusts. It must be calibrated first.
- Vector store: an index that returns approximate nearest neighbors for embedding queries.
- Workflow: a system in which LLMs and tools run through predefined code paths (Anthropic 2024 usage).

Cheatsheet

One page per theme. Formulas, counts, decision rules. All numbers from the course toys unless noted.

Core formulas

formula	meaning	where
$\text{pass}@k = 1 - (1-p)^k$	coverage: at least one of k passes	U01, U06
$\text{pass}^k = p^k$	reliability: all k pass	U06
gain per extra cost = $\frac{\text{delta}}{(\text{cost multiple} - 1)}$	ship rule vs the bar	U01, U04
$\text{SE}(\text{diff}) = \sqrt{p_1(1-p_1)/n + p_2(1-p_2)/n}$	noise on a comparison	U06

formula	meaning	where
$z = \text{delta}/\text{SE.}$	z	> 2 is ac- regression U06 tionable gate
LoRA params = $r(d + k)$	adapter count	U05
DPO loss = $-\log \text{sigma}(\text{beta } m)$	preference training	U05
$P(i \text{ beats } j) = \text{sigma}(s_i - s_j)$	Bradley-Terry	U06

formula	meaning	where
flat H steps = p^H	compounding checks	without U08
messages all-to-all = $n(n-1)$, hub = $2n$	coordination topology	U08
cost = auto share x auto cost + esc share x esc cost	FDE cost model	capstone B

Counts to memorize

- LoRA 1024x1024, $r = 8$: 16,384 trainable (64x fewer).
- 7B params: 14 GB fp16, 3.5 GB 4-bit.
- Grading per 1000: program \$1, model \$50, human \$2000.
- Timeout 0.1, $r = 3$: all-fail 0.001.
- $p = 0.3$, $k = 5$: pass@k 0.83, pass^k 0.0024.
- 1000 agents: 999,000 vs 2,000 messages per round.
- $0.99^{500} = 0.0066$.

Decision rules

decision	rule
workflow vs agent	path known: workflow. Steps unpredictable: agent
knob order	prompts, then weights, then compute
optimizer trust	gaps above 2 SE only
grader order	program, then model, then human sample

decision	rule
regression	block below -2 SE, celebrate above 2 SE, rerun otherwise
approval tier	auto (reversible), approve (costly), block (irreversible)
consent	implicit < explicit-once < explicit-each-time, by stakes

decision	rule
recovery	retry, rollback, escalate, compensate, cheapest first
eval tiers	dev (touch), test (schedule), sealed (once)
rollout	shadow, 10%, 50%, 100%, gates green 5 days each

Failure pairs (confused concepts)

- Grounding vs truth (U02).
- pass@k vs pass^k (U06).
- Retry vs idempotency (U03).
- Workflow vs agent (U04).
- Temperature vs top-k vs top-p (U01).
- Point vs pairwise judging (U06).
- Validator score vs human quality (U05).
- Demo vs production (U08).

Provenance shorthand

- S01-S04: released, schedule-title level.
- S05-S20, finals: PLANNED / SOURCE ATTRIBUTION PENDING.
- S10, S16 guests: TBA, logged in GAP-08.
- Everything else: independent theory, labeled.

Crash course

One-pass review of the full course. Each unit: the ideas, the key numbers, the decision rules. Detail lives in `lessons/u0N_lesson.md`.

U01: Compound systems and LLM foundations

A model is a function. A system is components with interfaces and per-interface failure modes. Decompose the task, then pick the architecture: the simple baseline first, the compound system only when the gain per cost clears the bar. Decoding: temperature, top-k, top-p are three different controls. Test-time scaling: $\text{pass@k} = 1 - (1 - p)^k$ measures coverage (at least one of k). Structured I/O plus constrained decoding make outputs machine-readable. Context engineering decides what fills the window.

Key numbers: pass@k at $p = 0.3$, $k = 8$: 0.94. Gain per extra cost = $\Delta / \text{cost} \times \text{multiple}$.

U02: Retrieval and evidence engineering

Grounding is not truth: a cited claim can still be false. The stack: chunk, embed, store, hybrid score (lexical + dense, normalized before fusion), rerank, cite, abstain. ColBERT keeps token vectors and scores with max-sim at query time. Cross-encoders read query and doc together (accurate, slow). Permissions filter before ranking, never after. Abstain when the evidence is insufficient.

Key numbers: fusion needs normalization (raw BM25 near 20 vs dense near 0.9). Leakage: ACLs applied after ranking leak snippets.

U03: Tools, protocols, and runtime safety

The tool/result loop is a state machine: thought, call, result. MCP: host/client/server, JSON-RPC 2.0, stdio (local trust) vs Streamable HTTP (network auth). Timeouts cut the tail. Retries need idempotency keys, or they double-charge. TTL must exceed the retry window. Approval tiers sort by reversibility. Sandboxes bound blast radius.

Key numbers: timeout 0.1, $r = 3$: all-fail 0.001. Zero doubles is the only passing count.

U04: Frameworks, workflows, memory, coordination

Five workflow patterns (chaining, routing, parallelization, orchestrator-workers, evaluator-optimizer) plus the agent: fixed code path vs model-directed. ReAct interleaves thought, action, observation. Memory: working context vs persistent store. Files for exact facts. Handoffs carry envelopes (task, state, constraints, done). Errors compound multiplicatively: $0.9^3 = 0.73$. Stopping rules: answer, budget, stall, denial.

Key numbers: gain per extra cost at bar 0.03 decides single vs multi.

U05: Optimization and agent data

Three knobs in spend order: prompts (words), weights (LoRA: $W + BA, r \ll d$), compute (bigger models, more calls). Prompt optimizers search wordings with sentence-shaped feedback. DPO trains on preference pairs with $-\log \sigma(\beta m)$, no RL. Data: traces (cheap, noisy), demos (expensive, clean), feedback (weak signal). Synthetic data needs an independent verifier. The flywheel compounds deployment data. Hygiene: quarantine train from eval, calibrate the validator before the optimizer trusts it, three tiers (dev/test/sealed).

Key numbers: LoRA 1024×1024 $r = 8$: 16,384 vs 1,048,576 (64x). Trust gaps above 2 SE only.

U06: Benchmark and evaluation infrastructure

A benchmark is a tuple: request, environment, stopping, scorer. No tuple, no benchmark. The scaffolding is part of the measurement (mock vs real: 0.13 gap on the toy). Graders: program (\$0.001), model (\$0.05), human (\$2). Cascade cheap-first. Judge prompts need criteria, anchors, examples, format. Point vs pairwise: Bradley-Terry turns wins into strengths. Judge biases: position, verbosity, self-preference. Swap and average, report seeds. $\text{pass}@k$ (coverage) vs pass^k (reliability). Isolate the test rig (fresh world per task). Inject faults to grade recovery. Tiny suites for speed, z-gates for regressions ($|z| > 2$), human alignment as the receipt.

Key numbers: $p = 0.3$, $k = 5$: $\text{pass}@k$ 0.83, pass^k 0.0024. Delta 0.02 on $n = 200$ is noise ($z = 0.4$).

U07: Safety, coding, and proactive agents

Privacy: minimize, log, audit. Instruction hierarchy: system > developer > user > tool output (data, never instructions). Tag sources. The planner obeys tags, not content. Red-team: attack, measure ASR, fix, re-attack. Permission boundary: capabilities enforced at the action layer. Approval tiers: auto/approve/block by reversibility. Coding agents: the test rig decides (tests), SWE-bench provenance (FAIL_TO_PASS + PASS_TO_PASS, strict). Checkpoints bound crash losses. Proactive: user model with confidence, next-action prediction above a precision gate, mixed initiative by confidence and stakes, consent gradient matching friction to risk.

Key numbers: ASR 0.24 to 0.06 ($z = 2.6$). Consent gradient: 10 min/day vs 50 flat.

U08: Frontiers, production, and project artifacts

Computer use: observe, ground, act, verify. Multimodality: ablate before you pay. Science agents: the oracle is the world. Cost per finding = $c/\text{hit-rate}$. Long horizon: planner/worker/checker. $0.99^{500} = 0.0066$ flat vs 0.90 with checks. Production: trace every span, attribute every dollar, p99 budgets, recovery ladder (retry, rollback, escalate, compensate), hub topology ($2n$ vs n^2), interpretability by counterfactual probe. Project: reviewable proposal (question, baseline, budget, metric, kill criterion), named owner, runbook, scoped capstone, honest gap log.

Key numbers: 1000 agents: 999,000 vs 2,000 messages per round. No trace, no production.

The twelve rules (one per course theme)

1. Name the interfaces before the system.
2. Baseline first, compound only on gain per cost.
3. Grounding is not truth.
4. Filter permissions before ranking.
5. Retries without idempotency double-charge.
6. Workflows when the path is known. Agents when it is not.
7. Spend words, then weights, then compute.
8. Calibrate the validator before the optimizer trusts it.
9. No tuple, no benchmark.

10. Data is never instructions.
11. Match the friction to the risk.
12. No trace, no production.